

JAVASCRIPT — LIVE CLOCK ON A WEBPAGE

STATIC VS LIVE CLOCKS: WHAT'S THE DIFFERENCE?



Look around you — clocks are everywhere!



Railway Station boards,



Mobile phones,



Cricket scoreboards — they all show time!

A Static Clock shows ONE fixed time and never changes.

Like a photo of a clock!

A Live Clock keeps updating every single second — tick, tick, tick! Just like your phone's clock.



Key Point: JavaScript helps us build LIVE clocks that update automatically every second!

MEET THE WEB DEVELOPMENT METRO TRAIN!

01

HTML = Iron body, tracks & doors 🚪

HTML gives the metro its structure — the coaches, doors, and tracks that hold everything together!

02

CSS = Paint, colours & route maps 🎨

CSS makes the metro look beautiful — bright colours, logos, and stylish route maps on the walls!

03

JavaScript = Announcements, doors & digital boards ⚡

JavaScript is the ACTION — auto doors, live display boards, and "Next Stop" announcements!

WHAT IS JAVASCRIPT? YOUR WEBSITE'S MAGIC ENGINE!

JavaScript is like the magic engine inside a website! Just like a train needs an engine to move, a website needs JavaScript to come alive and do fun things.

Without JavaScript, a webpage just sits there quietly. But with JavaScript, it can move, change, play sounds — and even show you the time every single second! JavaScript tells the clock when to tick! ⚡🕒

Every time you see something on a website that updates or reacts — that's JavaScript doing its magic!

THE MAGIC CAPSULE: DATE() OBJECT



Meet the Magic Capsule! ✨

JavaScript has a special magic capsule called `Date()` that holds today's time right inside it!

```
const indianTime = new Date();
```

When you write this, `indianTime` catches the current time from your computer's clock — like grabbing a snapshot of "right now"!

Just like a lunchbox holds your favourite food, the `Date()` capsule holds hours, minutes, and seconds.

Cool, right? 🕒

EXTRACTING TIME FROM THE MAGIC CAPSULE

01

getHours() — Opens the "Hours Dabba"!

Just like taking out the right spice from a kitchen dabba, `getHours()` reaches inside `indianTime` and pulls out the current hour. Example: `const currentHours = indianTime.getHours();`

02

getMinutes() — Opens the "Minutes Dabba"!

Now we open the minutes dabba! `getMinutes()` pulls out the current minutes from our magic capsule. Example: `const currentMinutes = indianTime.getMinutes();`

03

getSeconds() — Opens the "Seconds Dabba"!

The tiniest dabba holds the seconds! `getSeconds()` gives us the ticking seconds. Example: `const currentSeconds = indianTime.getSeconds();`

BUILDING YOUR MAGIC CLOCK FUNCTION: SHOWMYTIME()

A function is like a chai recipe — follow the steps and you get the same result every time! Our showMyTime() function has two simple steps to make your clock come alive on the webpage.

```
function showMyTime() {  
  // STEP 1: Get time  
  // STEP 2: Show time  
}
```

Can you guess what comes next? 🍵

01

Write the function skeleton:
function showMyTime() { } — this is
your recipe box, ready to be filled
with magic instructions!

02

STEP 1 — Open the magic capsule:
const indianTime = new Date(); —
grab the current time from your
computer's clock!

03

STEP 2 — Update the clock box on
your webpage so everyone can see
the time ticking live. Your recipe is
complete! 🎉

CONNECTING JAVASCRIPT TO YOUR WEBPAGE CLOCK BOX

Think of your HTML clock box like a student with an Identity Card (ID)! JavaScript uses `document.getElementById("clockBox")` to find and catch that exact box on the webpage.

Once JavaScript finds the box, it uses `innerText` to change what is written inside it — just like a teacher writing the time on a blackboard!

```
const clockBox = document.getElementById("clockBox");  
clockBox.innerText = currentHours + ":" + currentMinutes + ":" +  
currentSeconds;
```

Every second, JavaScript updates the clock box with the latest time. That's how your live clock stays fresh! 🕒

THE SCHOOL BELL MAGIC: USING SETINTERVAL()

`setInterval()` is like the school bell — it rings again and again, every second!

This magical instruction tells JavaScript to keep updating your clock:

```
setInterval(showMyTime, 1000);
```

1000 means 1000 milliseconds = 1 second!

Just like your school bell rings every period, `setInterval()` rings your clock's bell every single second — so your clock is always showing the right time! 🕒🔔

CUSTOMIZE YOUR CLOCK WITH GREETINGS!

Make your clock say hello like a friend! Add these lines
inside showMyTime():

```
if (currentHours < 12) {  
  greeting = "Good Morning, Beta!";  
} else if (currentHours < 18) {  
  greeting = "Good Afternoon, Beta!";  
} else {  
  greeting = "Good Evening, Beta!";  
}
```

```
clockBox.innerText = greeting + " " + currentHours + ":" +  
  currentMinutes + ":" + currentSeconds;
```

Now your clock greets you differently in the morning,
afternoon, and evening — just like a friend! 🌞🌤️🌙

MICRO-QUIZ: WHICH MAGICAL INSTRUCTION DID CHHOTU FORGET?

A

`stopClock()`

This instruction tells the clock to stop ticking. Would this help Chhotu's clock update every second? 🤔

B

`setInterval()` ✅ CORRECT ANSWER!

Chhotu forgot `setInterval(showMyTime, 1000)`! This is the school bell that rings every second to keep the clock updating!

C

`saveTime()`

This instruction does not exist in JavaScript magic! It would not help Chhotu's clock at all. 😊

ACHIEVEMENT BADGE CELEBRATION!



Congratulations! You've learned how to create a live clock using JavaScript magic.

You are now a JavaScript Clock Hero! Keep coding, keep creating — the web is yours to build! 🚀 ✨